

Artificial Intelligence and Machine Learning Challenges in Multimodal Diabetes Data: Wearables, Retinal Imaging, and Model Generalization

Saheed Tijani

Email: tijani.saheed@yahoo.com

Abstract

Background: Diabetes affects over 500 million people worldwide and strains healthcare systems. AI and machine learning (ML) show promise for prevention, diagnosis, and management, but combining wearable sensors, continuous glucose monitors, and retinal imaging creates technical and practical challenges that remain incompletely addressed.

Objective: This review examines key obstacles facing AI/ML in diabetes care: data inconsistency across devices and populations, poor model performance outside training conditions, unfair outcomes for certain patient groups, lack of transparency in model decisions, and deployment barriers in resource-limited settings. We pay particular attention to underserved communities and low- and middle-income countries, where the gap between research promise and clinical reality is widest.

Methods: We reviewed 361 papers published between 2018 and 2025 from SciSpace, Google Scholar, ArXiv, PubMed, and npj Digital Medicine. Our analysis covered the methods researchers use, the types of data they work with, how they combine multiple data sources, and what limitations they report.

Results: AI/ML systems perform well on standard test data but struggle on new populations, settings, or devices. Deep learning dominates but reproducibility is poor: studies rely on similar datasets, rarely use external validation, and omit key details. Multimodal fusion can improve accuracy but missing data and temporal alignment pose challenges. Models consistently perform worse for women, patients with poor glucose control, and underrepresented groups. In resource-limited settings, infrastructure, cost, privacy, and lack of local validation add further obstacles.

Conclusions: Equitable AI-driven diabetes care requires testing on diverse populations, reporting subgroup performance, using privacy-preserving methods like federated learning, and considering cost and infrastructure from the start. Without these changes, AI risks widening rather than closing health gaps.

Index Terms

Artificial Intelligence, Machine Learning, Diabetes, Multimodal Data, Wearable Sensors, Continuous Glucose Monitoring, Retinal Imaging, Model Generalization, Bias, Fairness, Low-Resource Settings

I. INTRODUCTION

Diabetes is a defining health crisis. The International Diabetes Federation reports 537 million adults with diabetes in 2021, rising to 783 million by 2045 [1], [2]. The disease consumes about 12% of global health spending and causes serious complications including heart disease, kidney failure, and blindness [2], [3]. Current management relies on periodic clinic visits, finger-prick glucose tests, and one-size-fits-all plans that struggle to account for the many factors affecting each patient [4], [5].

The rise of wearable technology and AI has opened new possibilities for diabetes care. Continuous glucose monitors now track blood sugar every few minutes, while smartwatches capture physical activity, heart rate, and sleep patterns [6], [7]. Retinal cameras can detect early signs of diabetic eye disease, which affects roughly one-third of people with diabetes and remains a leading cause of preventable blindness [8], [9]. Electronic health records contain years of lab results, prescriptions, and clinical notes [4]. Researchers have begun combining these data streams through AI systems that promise to predict glucose swings before they happen, catch complications earlier, and tailor treatments to individual patients [5], [10], [11].

Moving these systems from research into real clinics has proven difficult: models that excel on benchmarks often fail on new hospitals, populations, or devices [12]–[14]. Prioleau et al. [12] tested six glucose prediction models and found all made larger errors for women than men and worse for patients with poor glucose control. Wang et al. [15] documented similar disparities in retinal imaging for underrepresented groups. These biases reflect how data are collected, models trained, and success measured.

Low-resource settings sharpen these challenges. AI systems often assume reliable electricity, fast internet, and expensive devices—assumptions that do not hold in much of the world [5], [16], [17]. LMICs carry a disproportionate diabetes burden but remain almost invisible in AI training data [15], [18]; models built for well-resourced clinics often produce unreliable results when faced with different disease patterns, lower-quality images, or intermittent data [9], [17]. Federated learning [19], [20], transfer learning, and domain adaptation [21], [22] offer partial responses, and Bahmani et al. [23] propose integrating AI with education and capacity building—but these approaches remain largely experimental where they are most needed.

This review examines obstacles from data fusion to deployment in resource-limited settings, analyzes what current methods can and cannot do, identifies gaps in bias, generalization, and interpretability, and highlights directions for more accurate, fair, and accessible AI-driven diabetes care.

II. DATA MODALITIES AND PROBLEM FORMULATION

AI systems for diabetes care draw on several data types, each with distinct strengths and limitations.

A. Wearable Sensors and Continuous Glucose Monitoring

CGM has transformed blood sugar tracking. Devices measure interstitial glucose every 1–15 minutes, enabling detection of dangerous highs and lows, prediction of trends, and insulin fine-tuning [6], [21], [24]. Smartwatches and fitness trackers capture heart rate, activity, and sleep, which influence glucose regulation [6], [7]. Combining CGM with activity data allows models to adjust forecasts when someone has exercised or slept poorly [6], [21]. Heart-rate-based glucose estimation from wearables has not yet proven accurate enough for clinical use [6], [25].

Wearable data pose challenges: devices differ in sampling rates and measurement techniques, hindering cross-brand models [12], [26]; sensor dropout and gaps distort training and predictions [27], [28]; and calibration drift alters readings over time [6], [24]. Processing these high-frequency, noisy signals requires architectures like LSTMs and substantial compute and data [13], [14], [29].

B. Retinal Imaging Modalities

Diabetic retinopathy affects about 35% of patients and is a leading cause of blindness in working-age adults [8]; early screening can prevent vision loss but many patients lack access. Fundus photography and OCT are the main screening and diagnostic techniques [21].

AI-based retinal screening has achieved remarkable success in controlled settings. Fundus cameras capture color images of the back of the eye, revealing the tiny bleeds, protein deposits, and abnormal blood vessels that signal retinopathy [30], [31]. A systematic review of regulator-approved deep learning systems found pooled sensitivity of 93% and specificity of 90%, performance that matches or exceeds trained human graders [9], [32], [33]. OCT adds three-dimensional detail by imaging the retinal layers, helping clinicians assess swelling and structural damage [21], [34].

Real-world deployment remains difficult. Image quality varies with camera, operator, and patient factors; 3–12% of images are ungradable (higher for non-mydratic: 12.1% vs mydratic: 3.4%) [9], [35]. Models trained on research-grade images struggle with portable-camera quality in community settings [9], [18]. Most training data come from high-income countries [15], [17], and label noise arises from grader disagreement on borderline cases [9].

C. Electronic Health Records and Clinical Data

EHRs contain diagnoses, lab results, medications, vital signs, and notes [4], [21], enabling risk identification and earlier intervention [15]; NLP can extract structure from free text [21]. Challenges include high dimensionality, missing and irregular data [4], selection bias toward sicker patients with more complete records [24], and complex interactions that push toward deep learning [13].

D. Problem Formulation and Modeling Objectives

Researchers apply AI to diabetes care for several distinct purposes, each requiring different data, methods, and success metrics:

- **Glucose prediction:** Forecasting where blood sugar will be 30 to 60 minutes ahead, giving patients time to eat or take insulin before problems develop [12], [24].
- **Retinopathy screening:** Automatically grading eye images to determine which patients need specialist referral, enabling screening at scale [9], [13].
- **Risk prediction:** Estimating the likelihood that someone will develop diabetes or experience a complication like kidney disease or amputation [4], [15].
- **Treatment optimization:** Recommending personalized insulin doses, medication adjustments, or lifestyle changes [5], [21].

Each task requires careful choice of outcomes, prediction horizons, and clinical metrics [24]. Imbalanced complications can make standard accuracy misleading [14], and the progressive nature of diabetes demands modeling change over time [13].

Table I summarizes the key data types, their clinical uses, and the challenges each presents.

III. MACHINE LEARNING AND DEEP LEARNING APPROACHES

Diabetes AI has shifted from established statistical methods toward deep learning that extracts patterns from raw sensors and images; each approach has distinct trade-offs.

A. Traditional Machine Learning Methods

Logistic regression, SVMs, decision trees, and gradient boosting remain useful when data are limited or interpretability matters [14], [21]; for diabetes risk or short-horizon glucose forecasting, random forests and XGBoost often perform well [4], [14], [36]. Traditional methods run faster, need less data, handle missing values better, and offer interpretable feature importance [14]; Bayesian approaches add uncertainty estimates [21]. For structured clinical data they often match or beat neural networks [4]. The limitation appears with complex data: raw CGM, images, or free text require manual feature extraction that demands expertise and may miss clinically important patterns [13], [14], which drove the shift to deep learning.

B. Deep Learning Architectures for Time-Series Data

For glucose prediction from CGM data, recurrent neural networks have become the standard approach. LSTM networks and their relatives can learn patterns that unfold over hours, capturing how meals, insulin doses, and exercise affect blood sugar long after the event [12], [13], [37]. On benchmark datasets, these models achieve prediction errors below 20 mg/dL for 30-minute forecasts, accurate enough to be clinically useful [26], [29], [37].

Newer architectures have challenged LSTM dominance. Temporal convolutional networks process entire time series in parallel rather than step by step, making them faster to train [13]. Transformer models, borrowed from language processing, use attention mechanisms to focus on the most

relevant past measurements [14]. All achieve similar accuracy on standard tests [12].

Prioleau et al. [12] reproduced six glucose prediction models across three datasets and exposed a fundamental problem: models that worked on one dataset often failed on others despite similar patient and device characteristics. Errors were higher for women and for patients with poor glucose control—suggesting benchmark results and standard evaluations miss built-in biases.

C. Convolutional Neural Networks for Retinal Imaging

Deep convolutional networks have achieved striking success in automated retinal screening. Networks like ResNet, Inception, and EfficientNet can grade diabetic retinopathy from fundus photographs at accuracy levels matching specialist ophthalmologists [9], [13]. Transfer learning, where networks pre-trained on millions of general images are fine-tuned on medical data, has proven essential for reaching this performance with limited labeled examples [13].

Several strategies have improved results further. Combining multiple networks through ensembles reduces errors and improves calibration [30]. Attention mechanisms highlight which parts of an image drove the diagnosis, offering clinicians a window into the model’s reasoning [13], [38]. Training networks to detect multiple conditions simultaneously can improve learning efficiency and generalization [30].

Real-world deployment reveals gaps. Zhang et al. [9] found regulator-approved systems averaged 93% sensitivity and 90% specificity but performance varied with camera type, image quality, and population; models struggle with portable cameras in community screening [18], and demographic performance often goes unreported [15], [17]. Validated systems have nonetheless shown acceptable performance in diverse and low-resource settings [39]–[41].

D. Hybrid and Ensemble Approaches

No single method dominates across all diabetes AI tasks. Hybrid architectures that combine different approaches can capture both spatial and temporal patterns, such as CNN-LSTM networks that analyze sequences of retinal images or integrate imaging with time-series data [13], [14]. Ensembles that average predictions from multiple models tend to be more robust than any individual component [14], [36].

For clinical prediction tasks using structured EHR data, gradient boosting methods like XGBoost often outperform deep learning, especially when datasets are modest in size [4], [14]. These methods also provide interpretable outputs through feature importance rankings, helping clinicians understand which variables matter most [4]. The practical lesson is to match the method to the data: deep learning shines with images and raw signals, while tree-based methods may be better for tabular clinical data.

E. Emerging Paradigms: Transfer Learning and Federated Learning

Two methodological advances address persistent challenges in medical AI. Transfer learning tackles data scarcity by start-

TABLE I
COMPARISON OF DATA MODALITIES IN AI-DRIVEN DIABETES CARE

Data Modality	Key Characteristics	Clinical Applications	Major Challenges
Continuous Glucose Monitoring (CGM)	High-frequency time-series (1-15 min intervals); interstitial glucose	Glucose prediction, hypoglycemia detection, insulin optimization	Missing data, sensor noise, calibration drift, inter-device variability, high computational cost
Wearable Sensors (Activity, HR, PPG)	Multi-modal physiological signals; variable sampling rates	Context-aware glucose prediction, activity detection, stress monitoring	Heterogeneous devices, missing data, limited accuracy for glucose estimation, privacy concerns
Fundus Photography	2D color retinal images; variable quality	Diabetic retinopathy screening and grading	Image quality variability, limited diversity in training data, device heterogeneity, label noise
Optical Coherence Tomography (OCT)	High-resolution 3D retinal structure	Macular edema quantification, structural assessment	High cost, limited availability in low-resource settings, computational complexity
Electronic Health Records (EHR)	Longitudinal clinical data; structured and unstructured	Risk prediction, complication forecasting, treatment optimization	High dimensionality, sparsity, irregular sampling, non-random missingness, documentation bias
Genomic Data	Genetic variants, polygenic risk scores	Risk stratification, precision medicine	High cost, limited clinical utility, ethical concerns, population stratification

ing with models trained on large general datasets and adapting them to specific medical tasks [13], [21], [42]. A retinal screening network might begin with features learned from ImageNet, then refine them using diabetic retinopathy images. Domain adaptation techniques go further, explicitly adjusting for differences between training and target populations [19], [22].

Federated learning keeps data local while sharing only model parameters [19], [20]. Bai et al. [20] proposed a federated multimodal framework for precision-equitable diabetes care, enabling collaboration without raw data sharing and potentially including underrepresented populations; challenges remain in non-uniform data and fairness across sites [18], [19].

F. Reinforcement Learning for Treatment Optimization

Reinforcement learning offers an intriguing approach to insulin dosing. Rather than predicting outcomes, RL agents learn policies that map patient states to treatment actions, optimizing long-term glucose control while avoiding hypoglycemia [13], [21]. In simulation studies, these systems have learned effective dosing strategies without explicit programming of medical rules [21].

Deployment faces safety concerns [5]; offline RL from historical records avoids live experimentation but must evaluate actions never taken [21]. RL-based insulin optimization remains largely a research topic.

IV. MULTIMODAL LEARNING AND DATA FUSION

Diabetes leaves traces across blood sugar, retinal vessels, kidney function, and other measurements; multimodal AI can capture disease progression better than single data types. Lee et al. [11] and Singh et al. [43] showed that integrating EHR with wearable signals improves prediction over either alone, but technical challenges exceed those of single-modality approaches.

A. Fusion Strategies and Architectures

Fusion strategies include early fusion (merged features, one model; requires comparable scales [20]), late fusion (separate models, combined predictions; handles missing data but may miss cross-modal links [15]), and hybrid attention-based approaches that weight modalities by context [15], [21], [44]–[46]. Bai et al. [20] demonstrated a federated multimodal framework combining CGM and retinal data for complication risk with strong, decentralized results. Multimodal models demand more data, compute, and engineering; whether gains justify this complexity remains open.

B. Temporal Alignment and Missing Modalities

Different data types arrive on radically different schedules. CGM generates measurements every few minutes; retinal exams happen once a year; lab tests occur whenever a patient visits a clinic [4], [21]. Aligning these timescales requires deciding how to summarize continuous glucose data (daily averages? time in target range?) and how to handle the long gaps between imaging studies [13]. No consensus has emerged on the best approach.

Missing modalities are a larger problem: well-resourced patients may have CGM, eye exams, and full EHRs; rural clinics may offer only basic labs [5], [16]. Models expecting all modalities often fail when some are absent [15]. Modality dropout during training can help [20], but current methods perform best with everything and worst with least—the opposite of what equitable care needs.

C. Cross-Modal Transfer and Domain Adaptation

Transfer learning across modalities offers an intriguing possibility: could patterns learned from abundant retinal imaging data improve glucose prediction from scarce CGM records? The underlying idea is that different measurements capture related aspects of the same disease process [19], [21]. Retinal changes reflect long-term vascular damage; glucose patterns

reflect metabolic control. Linking them through shared learned representations might help each task [20].

Early results are promising but limited. Zhou et al. [47] showed that foundation models trained on diverse retinal images generalize better across populations than task-specific models. Chen et al. [22] demonstrated cross-modality knowledge transfer for retinopathy grading. However, the theoretical foundations of cross-modal transfer remain shaky, and negative transfer, where importing knowledge from one modality actually hurts performance on another, is a real risk [12], [15]. Careful validation is essential before assuming that connections between modalities will help.

D. Interpretability and Clinical Integration

Multimodal models compound interpretability: clinicians need to know which inputs drove a recommendation, but attention and saliency maps rarely capture full cross-modal logic [5], [13], [24]. Models that require all modalities cannot serve patients with partial data [16]; ideal systems would improve progressively as more data arrive [3], [5], [15]. Hierarchical decision support remains a research challenge.

E. Evaluation Challenges and Reporting Standards

Evaluating multimodal systems demands more than reporting overall accuracy. Researchers must assess robustness to missing inputs, fairness across patient subgroups, and calibration of confidence estimates [12], [15]. Ablation studies that systematically remove each modality reveal which data sources actually matter and whether seemingly complementary inputs are redundant [24].

Practice falls short: heterogeneous preprocessing, splits, and outcomes make comparison nearly impossible [12], [24]; demographic reporting is often absent [15]; and no widely accepted multimodal diabetes benchmark exists [12].

Figure 1 illustrates a conceptual framework for multimodal AI in diabetes care, showing how diverse data sources flow through specialized processors before fusion and clinical decision support.

V. MODEL LIMITATIONS: BIAS, GENERALIZATION, AND INTERPRETABILITY

Benchmark results tell only part of the story. This section examines three persistent challenges: systematic bias, poor performance under distribution shift, and lack of interpretability.

A. Algorithmic Bias and Fairness

Systematic performance gaps across demographic groups are among the most troubling findings in diabetes AI [12], [15]. Prioleau et al. [12] found all six glucose prediction models made larger errors for women than men across three cohorts and multiple architectures—pointing to how systems are built and evaluated. Retinal screening shows similar disparities: training data mostly from high-income countries, variable performance in underrepresented populations [9], [15], [17].

Phene et al. [39] in Zambia and Wolf et al. [40] in diverse pediatric populations show bias is not inevitable with deliberate validation.

Perhaps most troubling is the pattern Prioleau et al. [12] documented: models consistently underperform for patients with poor glucose control. These are precisely the patients who need the most help. If AI systems work best for people who are already doing well, they risk widening the gap between those who manage their diabetes effectively and those who struggle [48], [49].

Bias stems from overrepresented training data, selection toward patients with better access, accuracy-focused optimization that sacrifices minority subgroups, and metrics that average across patients [4], [15], [19]. Addressing it requires diverse data with explicit targets [19], [20], fairness-aware algorithms (with accuracy trade-offs [19]), and standard fairness audits by demographics and severity [15], [49].

B. Generalization and Dataset Shift

Models that perform well on their test set often fail in practice when deployment conditions differ from development [12], [13], [47]. Prioleau et al. [12] found poor conceptual reproducibility when applying glucose models across datasets—same code and device type, but different titration, diet, and activity; models learned non-transferable, site-specific patterns. Device differences compound this: CGM sensors differ in noise, sampling, and calibration [6], [12], [27]; research-grade and portable retinal cameras produce very different images [9], [18]. Most studies use same-source held-out data [12], [24]; independent validation consistently shows degradation [12], [15]. Improving generalization requires domain adaptation [21], [22], multi-site or federated training [18]–[20], and mandatory external validation [12], [15].

C. Interpretability and Explainability

Clinicians need to know why a model recommends an action [5], [24]; for safety-critical decisions, unexplained predictions are hard to trust or debug [5], [21]. SHAP, attention weights, and saliency maps offer partial views [13], [14], [38], [50]–[52] but are unstable, non-causal, or artifact-prone [5], [13], [14], [24]. Inherently interpretable models (trees, rules, linear) trade accuracy [14], [21]; hybrid interpretable–opaque designs remain challenging [13]. Regulators emphasize transparency and oversight [5], [10] but there is no consensus on adequate explanation or interpretability metrics [24].

D. Reproducibility and Reporting Standards

Prioleau et al. [12] reviewed 60 glucose-prediction papers and found code rarely available, methods incomplete, and results only on familiar benchmarks; heterogeneous preprocessing, metrics, and splits make comparison nearly impossible [24]. Demographic reporting is often absent [12], [15]. TRIPOD and CONSORT-AI offer reporting standards [15], [49]; code and data sharing and standard benchmarks with fixed protocols would improve reproducibility and comparability [12], [24].

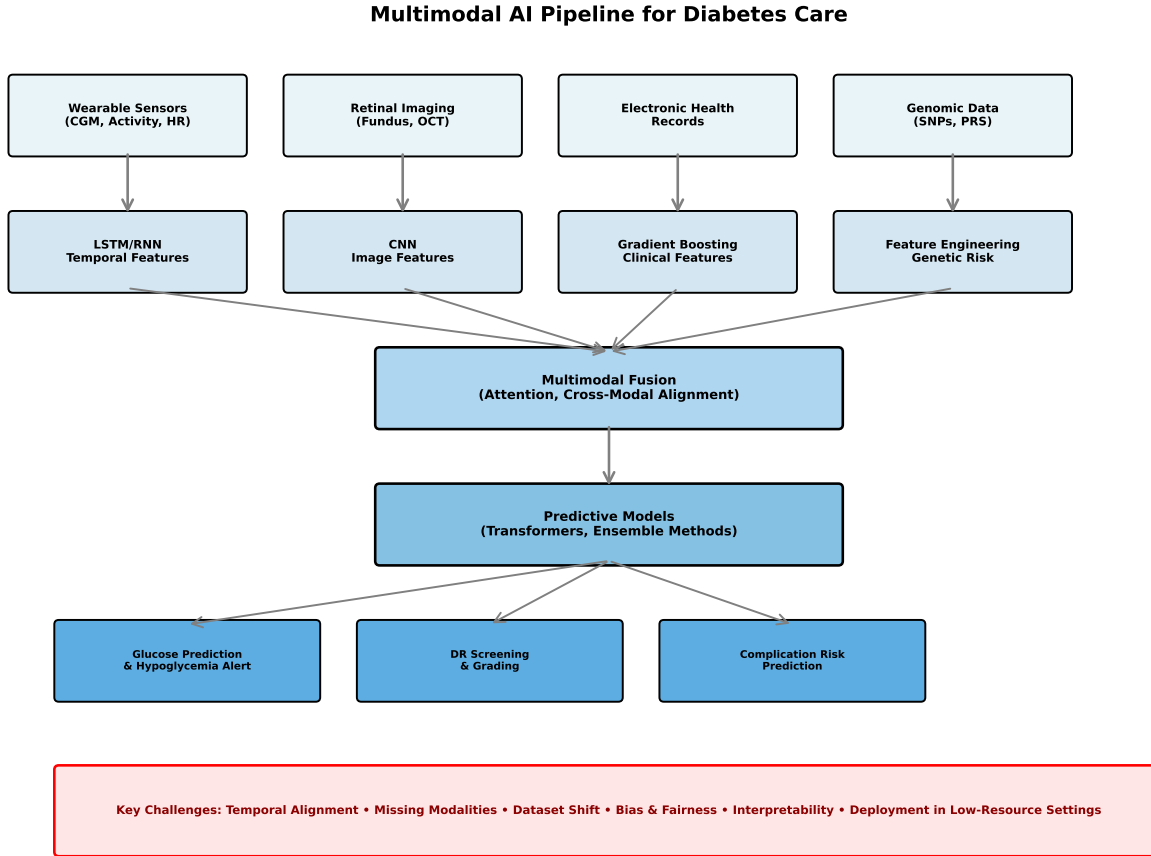


Fig. 1. Conceptual framework for multimodal AI in diabetes care. The pipeline integrates heterogeneous data sources (wearable sensors, retinal imaging, EHR, genomic data) through modality-specific feature extraction, cross-modal fusion, and predictive modeling to support clinical decision-making. Key challenges include temporal alignment, missing modalities, interpretability, and generalization across populations and clinical settings.

VI. DEPLOYMENT AND ACCESSIBILITY CHALLENGES IN LOW-RESOURCE SETTINGS

Deploying AI in clinical practice is difficult everywhere; obstacles multiply in resource-limited settings. LMICs bear a disproportionate diabetes burden yet receive little attention in AI development [1], [17].

A. Infrastructure and Connectivity Constraints

Most AI systems assume stable internet, reliable electricity, and computing power [5], [16], [53]; Eze et al. [54] found infrastructure failures among the most common reasons digital health interventions fail in LMICs. Paper records remain standard in many facilities [16], [18]; where EHRs exist they often lack standardized coding, identifiers, and interoperability [5], [17]. Mobile apps can deliver decision support without extensive IT [7], [14], [16] but are not universal and require digital literacy [5].

B. Device Costs and Economic Barriers

The devices that generate data for AI systems remain far too expensive for most patients in LMICs. CGM systems cost \$200-400 per month in high-income countries, an impossible

expense where annual health spending per person is often below \$100 [2], [16]. Even basic glucometers and test strips create ongoing costs that many families cannot sustain [6].

Retinal imaging faces similar economics. Desktop fundus cameras typically cost \$10,000-16,000, while portable handheld alternatives range from \$3,000-9,000 [9], [18]. The cheaper devices produce lower-quality images, which degrades AI performance, creating a tradeoff between accessibility and accuracy [9]. The shortage of trained camera operators and ophthalmologists compounds the problem [3], [9].

AI may reduce long-term costs by preventing complications, but upfront investment is often prohibitive [3], [5]; creative financing and technology transfer will be needed [16].

C. Data Scarcity and Population Representativeness

AI models learn from data, and the data overwhelmingly comes from wealthy countries. Raj et al. [18] and Olusanya et al. [17] document the severe underrepresentation of LMIC populations in publicly available training sets. Models optimized for European or North American patients may not work well for people with different genetics, diets, and disease patterns [9], [15].

DR prevalence and progression differ across populations [8], [9]; diet and lifestyle vary, so models trained on Western data may fail elsewhere [16]. Local validation is essential but resources are scarce. Transfer learning [21], [22] and federated learning [19], [20] offer paths forward; Phene et al. [39] showed a Singapore-trained system performed well in Zambia. Negative transfer—assumptions from one setting hurting another—remains a risk [12].

D. Clinical Validation and Regulatory Frameworks

Trials in low-resource settings face infrastructure, funding, and expertise hurdles [5], [19]; pragmatic implementation studies help but rarely establish causality [5]. Many LMICs lack AI medical device guidelines [5], [10]; well-resource-oriented regulations can be impractical elsewhere [16]. Evidence from high-income settings does not automatically apply [19], [55]; local validation is essential and building that capacity is a major undertaking [5].

E. Workforce Capacity and Training

Specialists who could oversee AI-assisted care—endocrinologists, ophthalmologists, data scientists—are in short supply in many LMICs [3], [16], [18]. Primary care delivers most diabetes care but may lack training or confidence to integrate AI [5]; hesitancy over accuracy, liability, and workflow is legitimate [3]. Trust requires hands-on training, transparent communication, and evidence of benefit [5], [24]. Bahmani et al. [23] propose integrating research, education, and practice to democratize AI knowledge; technology alone cannot achieve equity without trained people and strong institutions [5], [23].

F. Ethical and Socio-Cultural Considerations

Systems built in high-income settings may not address local priorities or values; deployment without local validation risks imposing solutions rather than developing in partnership [5], [16]. Privacy concerns are heightened where laws and security capacity are weak [5], [10]; consent and scrutiny of commercial interests matter [19]. Community engagement can align AI with local needs [5]. Cultural factors—local foods, religious practices, language, literacy—must shape recommendations and interfaces [5], [16].

G. Pathways to Equitable AI Deployment

Making AI work in low-resource settings requires deliberate effort on multiple fronts:

- **Technology adaptation:** Design devices and algorithms for resource-constrained environments, including edge computing that works without constant internet and models that remain useful with lower-quality input data [16], [19], [56].
- **Capacity building:** Invest in training healthcare workers, researchers, and data scientists who can adapt, validate, and maintain AI systems locally [5], [23].
- **Collaborative research:** Use federated learning and multi-site partnerships to include diverse populations in

model development while respecting data sovereignty and privacy [18]–[20].

- **Inclusive design:** Involve patients, clinicians, and communities in developing systems to ensure cultural appropriateness and practical usability [5], [23].
- **Policy innovation:** Create regulatory frameworks and financing mechanisms suited to local contexts, balancing safety requirements with the need for accessible technology [5], [10].

Without attention to these priorities, the benefits of AI in diabetes care will accrue primarily to those who already have good access to healthcare, widening rather than narrowing global health disparities [15], [19], [48].

VII. OPEN RESEARCH GAPS AND FUTURE DIRECTIONS

The challenges above point to specific research priorities in methods, data, validation, and policy.

A. Methodological Advances

1) *Robust and Generalizable Models:* Domain adaptation that accounts for source–target differences shows promise but is underexplored in diabetes [19], [22]; meta-learning could enable adaptation with minimal new data [13]. Causal inference may improve generalization [4], [24] but applying it to high-dimensional time series and images remains challenging.

2) *Fairness-Aware Learning:* Fairness-aware training [15], [19] involves trade-offs (e.g., improving one group may harm another [19]); choosing acceptable trade-offs requires ethicists, clinicians, and communities [5], [49]. Intersectional fairness is harder: a model can do well for women and for Black patients on average yet fail for Black women [15], [48], and methods and data for small subgroups are lacking.

3) *Uncertainty Quantification and Calibration:* Uncertainty quantification (Bayesian deep learning, ensembles, conformal prediction) remains limited in diabetes AI [21], [52], [57]; calibration is often poor for rare events and unusual patients [15], [49]. Models should express greater uncertainty on out-of-distribution data [12], [24]; reliable quantification under shift is essential for safe deployment.

4) *Interpretable and Explainable AI:* Interpretable-by-design models (neural additive, concept bottleneck, prototype-based) and hybrid interpretable–opaque architectures are promising [13], [14], [38]. The field needs standardized interpretability metrics and human-centered studies of whether explanations help clinicians [5], [24].

B. Data and Infrastructure

1) *Diverse and Representative Datasets:* Diverse training data from underrepresented populations, multiple devices, and varied settings is essential [12], [15], [20] and requires addressing privacy, consent, and funding barriers [5]. Federated learning can enable global collaboration without centralizing data [18]–[20]; challenges in efficiency, heterogeneity, and fairness across sites need further work [19].

2) *Standardized Benchmarks and Evaluation Protocols:*

The field needs agreed benchmarks with fixed splits, standard metrics, and required reporting—including diverse populations and evaluation beyond aggregate accuracy (subgroups, calibration, robustness) [12], [15], [24]. Mandatory reporting of demographics, preprocessing, and hyperparameters would improve reproducibility [12].

C. *Clinical Translation and Implementation*

1) *Prospective Clinical Trials:* Most work is retrospective; prospective trials are needed to show AI improves outcomes [5], [19]. Wolf et al.'s ACCESS trial [40] demonstrated one model. Trials should include diverse and low-resource settings [5], [16]; pragmatic effectiveness designs are valuable for implementation [3].

2) *Human-AI Collaboration:* Human-AI collaboration—interface design, workflow, when to override, preserving skills and autonomy—remains poorly understood [5], [24].

3) *Health Economics and Cost-Effectiveness:* Resource-constrained health systems need evidence about whether AI tools are worth the investment. Rigorous health economic evaluations must account for device and infrastructure costs, ongoing operational expenses, and long-term savings from prevented complications [3], [5], [16]. Budget impact analyses are essential for low-resource contexts where affordability determines adoption.

D. *Regulatory and Policy Innovation*

1) *Adaptive Regulatory Frameworks:* Existing medical device regulations assume static technologies and struggle with AI systems that learn and change over time [5], [10]. New frameworks are needed that allow iterative updates while maintaining safety oversight. Risk-based approaches that tailor requirements to clinical impact could balance innovation with patient protection [10].

International harmonization could reduce duplicative approval processes and accelerate global access, though local adaptation may be necessary to account for different healthcare contexts [5], [16].

2) *Data Governance and Privacy:* Enabling research while protecting patients requires better governance frameworks. Federated learning, differential privacy, and secure computation offer technical solutions, but legal and institutional barriers persist [5], [19], [20]. Participatory models that involve patients and communities in decisions about data use may build trust and ensure alignment with stakeholder values [5].

E. *Education and Capacity Building*

AI education in curricula and capacity building in LMICs—where need is greatest and expertise most limited—are essential [5], [16], [19], [23]. Bahmani et al. [23] propose integrating education, research, and practice; international partnerships can support local development and validation [5].

F. *Emerging Technologies*

1) *Digital Twins and Personalized Simulation:* Digital twins create virtual patient representations for prediction and treatment optimization [55]; integrating CGM, wearables, and clinical history could enable simulated treatment response and dosing optimization. Validation and generalization to diverse populations need further work.

2) *Large Language Models:* Large language models could support education, self-management, and documentation, especially where healthcare workers are scarce [5], [16], [58]; accuracy, bias, and safety require careful validation before deployment [5].

3) *Advanced Sensors:* Non-invasive and multi-analyte sensors could expand continuous monitoring and enable AI-driven closed-loop systems [6], [7]; accuracy, calibration, biocompatibility, and cost remain barriers [5], [6].

VIII. CONCLUSION

AI offers real promise: deep learning can match specialists at retinopathy detection, predict glucose swings, and integrate data at scale [9], [11], [37]; several tools have regulatory approval and trials show benefit [40], [41]. This review also documents a substantial gap between benchmarks and clinical reality. Models that excel on test data routinely fail on new populations, devices, or settings. Prioleau et al. [12] found poor conceptual reproducibility across datasets and worse performance for women and for patients with poor glucose control—those who need help most. Retinal screening shows similar variability [9], [17].

These reflect how research is conducted: overrepresented training data, metrics that hide subgroup disparities, and validation rarely outside original context [12], [15]. Low-resource settings expose the weaknesses: infrastructure, cost, data scarcity, and workforce barriers [16], [53], [54]; LMICs bear a disproportionate burden but remain nearly invisible in training data [1], [17], [18]. Without deliberate inclusion, AI risks concentrating benefits among those already well-served [15], [48].

The path forward requires diverse data with explicit targets [19], [20], fairness and domain adaptation in model development [19], [22], and evaluation that includes subgroups, external validation, calibration, and uncertainty [12], [24], [49]. Federated learning can include underrepresented populations without sharing raw data [18], [20]; transfer learning and domain adaptation can help where local data are scarce, with validation to avoid negative transfer [22], [47]. Regulatory frameworks, education, and economic evidence for resource-constrained settings are needed [3], [5], [10], [16], [23].

The measure of AI in diabetes care should be whether it improves outcomes for people who need help: systems reliable across populations and settings, fair across demographics and disease severity, interpretable, and accessible where resources are limited. Diabetes affects hundreds of millions and will affect more [1]. Realizing AI's potential requires prioritizing equity and robustness alongside performance and serving all who need care, not only those easiest to reach.

REFERENCES

- [1] H. Sun, P. Saeedi, S. Karuranga, M. Pinkepank, K. Ogurtsova, B. B. Duncan, C. Stein, A. Basit, J. C. N. Chan, J. C. Mbanya, M. E. Pavkov, A. Ramachandran, S. H. Wild, S. James, W. H. Herman, P. Zhang, C. Bommer, S. Kuo, E. J. Boyko, and D. J. Magliano, "IDF Diabetes Atlas: Global, regional and country-level diabetes prevalence estimates for 2021 and projections for 2045," *Diabetes Research and Clinical Practice*, vol. 183, p. 109119, 2022.
- [2] International Diabetes Federation, *IDF Diabetes Atlas*, 10th ed. Brussels: International Diabetes Federation, 2021. [Online]. Available: <https://diabetesatlas.org>
- [3] B. Guan, J. Bhattacharya, H. Xiao, Y. C. Kudva, R. Basu, P. Herrero, and P. Georgiou, "Artificial intelligence in diabetes management: advancements, opportunities, and challenges," *Cell Reports Medicine*, vol. 4, no. 11, p. 101213, 2023.
- [4] E. K. Oikonomou and R. Khera, "Machine learning in precision diabetes care and cardiovascular risk prediction," *Cardiovascular Diabetology*, vol. 22, p. 259, 2023.
- [5] I. S. Mackenzie, I. Ford, A. Walker, N. Munro, G. Guthrie, G. P. Leese, R. S. Lindsay, J. A. McKnight, A. D. Morris, E. R. Pearson, J. R. Petrie, S. Philip, S. H. Wild, and N. Sattar, "Diabetes and artificial intelligence beyond the closed loop: a review of the landscape, promise and challenges," *Diabetologia*, vol. 66, pp. 1597–1612, 2023.
- [6] A. Y. Alhaddad, H. Aly, H. Gad, A. Al-Ali, K. K. Sadasivuni, J.-J. Cabibihan, and R. A. Malik, "Sense and learn: Recent advances in wearable sensing and machine learning for blood glucose monitoring and trend-detection," *Frontiers in Bioengineering and Biotechnology*, vol. 10, p. 876672, 2022.
- [7] I. Rodríguez-Rodríguez, J.-V. Rodríguez, I. Chatzigiannakis, and M. Zamora Izquierdo, "Applications of the internet of medical things to type 1 diabetes mellitus," *Electronics*, vol. 12, no. 3, p. 756, 2023.
- [8] J. W. Y. Yau, S. L. Rogers, R. Kawasaki, E. L. Lamoureux, J. W. Kowalski, T. Bek, S.-J. Chen, J. M. Dekker, A. Fletcher, J. Grauslund, S. Haffner, R. F. Hamman, M. K. Ikram, T. Kayama, B. E. K. Klein, R. Klein, S. Krishnaiah, K. Mayber, J. P. O'Hare, T. J. Orchard, M. Porta, M. Rema, M. S. Roy, T. Sharma, J. Shaw, H. Taylor, J. M. Tielsch, R. Varma, J. J. Wang, N. Wang, S. West, L. Xu, M. Yasuda, X. Zhang, P. Mitchell, and T. Y. Wong, "Global prevalence and major risk factors of diabetic retinopathy," *Diabetes Care*, vol. 35, no. 3, pp. 556–564, 2012.
- [9] J. Zhang, Y. Zheng, D. S. W. Ting, and T. Y. Wong, "Systematic review and meta-analysis of regulator-approved deep learning systems for fundus diabetic retinopathy detections," *npj Digital Medicine*, vol. 8, p. 12, 2025.
- [10] M. Khalifa and M. Albadawy, "Artificial intelligence for diabetes: Enhancing prevention, diagnosis, and effective management," *Computer Methods and Programs in Biomedicine Update*, vol. 5, p. 100141, 2024.
- [11] S. Lee *et al.*, "Large language multimodal models for new-onset type 2 diabetes prediction using five-year cohort electronic health records," *Scientific Reports*, vol. 14, p. 20123, 2024.
- [12] T. Prioleau, J. Moore, R. J. Galindo, and R. K. A. Basu, "Deep learning for blood glucose prediction: Reproducibility challenges and factors affecting differential performance," *Research Square*, 2025.
- [13] T. Zhu, K. Li, P. Herrero, and P. Georgiou, "Deep learning for diabetes: A systematic review," *IEEE Journal of Biomedical and Health Informatics*, vol. 25, no. 7, pp. 2744–2757, 2021.
- [14] M. T. Alam, M. M. Hasan, M. A. Alam, A. S. Nitu, M. P. Hossain, and M. Z. H. Alam, "Machine learning and artificial intelligence in diabetes prediction and management: A comprehensive review of models," *Journal of Novel Engineering Science and Technology*, vol. 1, no. 1, pp. 41–52, 2024.
- [15] J. Wang, Y. Huang, S. Jabbour, S. Mukherjee, Z. Neshati, S. D. Young, E. Lehman, J. Kalpathy-Cramer, L. A. Celi, and L. G. McCoy, "Ai-based diabetes care: risk prediction models and implementation concerns," *npj Digital Medicine*, vol. 7, p. 34, 2024.
- [16] S. Ghosh, A. Dasgupta, and A. Swetapadma, "Artificial intelligence in personalized medicine for diabetes mellitus: a narrative review," *Cureus*, vol. 17, no. 1, p. e91520, 2025.
- [17] B. O. Olusanya, D. J. Weiss *et al.*, "Mitigating machine learning bias between high income and low-middle income countries for enhanced model fairness and generalizability," *Scientific Reports*, vol. 14, p. 13318, 2024.
- [18] A. Raj, J. Saini, and S. Bhatia, "Federated learning for diabetic retinopathy diagnosis: Enhancing accuracy and generalizability in under-resourced regions," *arXiv preprint*, 2024, arXiv:2411.00869.
- [19] H. Fahmy, "Exploring the role of ai in predicting chronic disease progression: Diabetes and cardiovascular diseases," *Premier Journal of Public Health*, vol. 1, no. 1, p. 21, 2025.
- [20] J. Bai, T. Zhu, P. Herrero, and P. Georgiou, "Federated multimodal ai for precision-equitable diabetes care," *Frontiers in Digital Health*, vol. 6, p. 1678047, 2025.
- [21] I. Contreras and J. Vehi, "Artificial intelligence for diabetes management and decision support: Literature review," *Journal of Medical Internet Research*, vol. 20, no. 5, p. e10775, 2018.
- [22] X. Chen *et al.*, "Cross-modality transfer learning with knowledge infusion for diabetic retinopathy grading," *Frontiers in Medicine*, vol. 11, p. 1400137, 2024.
- [23] A. Bahmani, K. Asare, L. E. Boulware, L. Brewer, A. J. Butte, D. Char, R. Chetty, K. Churchwell, L. Cipriano, M. Desai, S. N. Duda, J. C. Eichstaedt, E. Eisenstein, L. Flowers, J. Gichoya, S. L. Gómez, K. Goodman, L. Gottlieb, R. Hahn, T. Hernandez-Boussard, Y. Hsuen, A. Kaushal, I. S. Kohane, C. R. Lyles, M. E. Matheny, K. Morse, E. Nsoesie, Z. Obermeyer, J. A. Omiye, J. Pathak, S. R. Pfohl, P. Rajpurkar, D. Rehkopf, F. Rodriguez, L. G. Rosas, A. Rusanov, A. Schuler, N. H. Shah, K. Shankar, M. Sjoding, S. Saria, T. Veinot, A. Verghese, J. Voigt, M. Weiner, D. R. Williams, C. Wong, J. R. Zubizarreta, T. Hernandez-Boussard, and N. H. Shah, "Achieving inclusive healthcare through integrating education and research with ai and personalized curricula," *Communications Medicine*, vol. 5, p. 34, 2025.
- [24] P. G. Jacobs, J. E. Youssef, and J. R. Castle, "Artificial intelligence and machine learning for improving glycemic control in diabetes: best practices, pitfalls, and opportunities," *IEEE Reviews in Biomedical Engineering*, vol. 17, pp. 83–101, 2023.
- [25] S. Zanelli, E. Yacoub, A. Marzullo, C. Corsi, and B. De Maria, "Diabetes detection and management through photoplethysmographic and electrocardiographic signals analysis: A systematic review," *Sensors*, vol. 22, no. 13, p. 4890, 2022.
- [26] S. Langarica, M. Munoz-Organero, and G. Ramírez-González, "Deep learning-based glucose prediction models: A guide for practitioners and a curated dataset for improved diabetes management," *IEEE Open Journal of Engineering in Medicine and Biology*, vol. 5, pp. 176–187, 2024.
- [27] F. Xie *et al.*, "A processing algorithm to address real-world data quality issues with continuous glucose monitoring data," *Journal of Diabetes Science and Technology*, 2024.
- [28] I. B. Hirsch *et al.*, "Interruption of continuous glucose monitoring: Frequency and adverse consequences," *Journal of Diabetes Science and Technology*, vol. 18, no. 5, pp. 1166–1173, 2024.
- [29] J. Martinsson, A. Schliep, B. Eliasson, C. Meijner, S. Persson, and O. Mogren, "Blood glucose prediction with variance estimation using recurrent neural networks," *Journal of Healthcare Informatics Research*, vol. 4, no. 1, pp. 1–18, 2020.
- [30] G. Selvachandran, S. G. Quek, R. Paramesran, W. Ding, and L. H. Son, "Developments in the detection of diabetic retinopathy: a state-of-the-art review of computer-aided diagnosis and machine learning methods," *Artificial Intelligence Review*, vol. 56, pp. 915–964, 2022.
- [31] H. AbuShawish, E. Alajrami, and S. S. Abu-Naser, "Deep learning in automatic diabetic retinopathy detection and grading systems: A comprehensive survey and comparison of methods," *IEEE Access*, vol. 12, pp. 84 883–84 906, 2024.

- [32] Y. Liu *et al.*, "Performance of artificial intelligence in diabetic retinopathy screening: a systematic review and meta-analysis of prospective studies," *Frontiers in Endocrinology*, vol. 14, p. 1197783, 2023.
- [33] J.-Y. Wu *et al.*, "The efficacy of artificial intelligence in diabetic retinopathy screening: a systematic review and meta-analysis," *BMC Ophthalmology*, vol. 23, p. 525, 2023.
- [34] S. Padhy *et al.*, "A systematic review on diabetic retinopathy detection and classification based on deep learning techniques using fundus images," *PeerJ Computer Science*, vol. 10, p. e1947, 2024.
- [35] P. Kanclerz, R. Tuuminen, and R. Khoramnia, "Imaging modalities employed in diabetic retinopathy screening: A review and meta-analysis," *Diagnostics*, vol. 11, no. 10, p. 1802, 2021.
- [36] M. T. Alam, M. M. Hasan, M. A. Alam, A. S. Nitu, M. P. Hossain, and M. Z. H. Alam, "Machine learning and artificial intelligence in diabetes prediction and management: a comprehensive review of models," *Social Science Research Network*, 2025.
- [37] Q. Sun *et al.*, "Long-term prediction of blood glucose levels in type 1 diabetes using a cnn-lstm-based deep neural network," *Journal of Diabetes Science and Technology*, vol. 16, no. 6, pp. 1474–1483, 2022.
- [38] A. Bhati, N. Gour, P. Khanna, and A. Ojha, "An interpretable dual attention network for diabetic retinopathy grading: Idanet," *Artificial Intelligence in Medicine*, vol. 150, p. 102782, 2024.
- [39] S. Phene, R. C. Dunn, N. Hammel, Y. Liu, J. Krause, N. Kitade, M. Schaekermann, R. Sayres, D. J. Wu, A. Bora, C. Semturs, A. Mishra, A. Krishnan, H. Nishikawa, K. Widner, G. Corrado, L. Peng, and D. R. Webster, "Artificial intelligence using deep learning to screen for referable and vision-threatening diabetic retinopathy in africa: a clinical validation study," *The Lancet Digital Health*, vol. 1, no. 1, pp. e35–e44, 2019.
- [40] R. M. Wolf *et al.*, "Autonomous artificial intelligence increases screening and follow-up for diabetic retinopathy in youth: the access randomized control trial," *Nature Communications*, vol. 15, p. 421, 2024.
- [41] M.-C. Boucher *et al.*, "Implementation of artificial intelligence-based diabetic retinopathy screening in a tertiary care hospital in quebec: Prospective validation study," *JMIR Diabetes*, vol. 9, p. e59867, 2024.
- [42] A. Ran, S. Niu, H. He, and Y. Zheng, "Source-free active domain adaptation for diabetic retinopathy grading based on ultra-wide-field fundus images," *Computers in Biology and Medicine*, vol. 174, p. 108418, 2024.
- [43] A. Singh *et al.*, "Medfusionai: A deep learning framework for multi-modal health data fusion to predict chronic disease risks," *International Journal of Computational and Experimental Science and Engineering*, vol. 10, p. 2159, 2024.
- [44] S. Kulkarni and B. E. Reddy, "Diabetes detection and prediction through a multimodal artificial intelligence framework," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 38, no. 1, pp. 459–468, 2025.
- [45] S. Maqsood, S. Xu, M. Springer, N. Mohammadzadeh, M. K. Rutter, I. Buchan, and K. Dashtipour, "Gluconet-mm: A multimodal attention-based multi-task learning framework with decision transformer for personalised and explainable blood glucose forecasting," *Diabetes, Obesity and Metabolism*, vol. 27, no. 3, pp. 1163–1176, 2025.
- [46] Y. Xiong, W. Zhang, and J. Chen, "A multi-modal deep learning approach for predicting type 2 diabetes complications: Early warning system design and implementation," *World Journal of Innovation and Modern Technology*, vol. 7, no. 6, p. 15, 2024.
- [47] Y. Zhou *et al.*, "Towards population-independent, multi-disease detection in fundus photographs," *Scientific Reports*, vol. 13, p. 11493, 2023.
- [48] S. A. Haider, S. Bornha, C. A. Gomez-Cabello, S. M. Pressman, C. R. Haider, and A. J. Forte, "The algorithmic divide: A systematic review on ai-driven racial disparities in healthcare," *Journal of Racial and Ethnic Health Disparities*, 2024, pMID: 39695057.
- [49] X. Liu *et al.*, "A scoping review and evidence gap analysis of clinical ai fairness," *npj Digital Medicine*, vol. 8, p. 45, 2025.
- [50] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, pp. 4765–4774, 2017.
- [51] M. Munoz-Organero *et al.*, "The importance of interpreting machine learning models for blood glucose prediction in diabetes: an analysis using shap," *Scientific Reports*, vol. 13, p. 16865, 2023.
- [52] K. Chadaga *et al.*, "Interpretable machine learning framework for diabetes prediction: Integrating smote balancing with shap explainability for clinical decision support," *Healthcare*, vol. 13, no. 20, p. 2588, 2024.
- [53] H. Desai and T. Katapally, "Barriers and facilitators to the implementation of artificial intelligence enabled diabetes interventions in lower-middle-income countries: A systematic review protocol," *medRxiv*, 2025.
- [54] N. D. Eze *et al.*, "Barriers and facilitators for the sustainability of digital health interventions in low and middle-income countries: A systematic review," *Frontiers in Digital Health*, vol. 4, p. 1014375, 2022.
- [55] G. Cappon and A. Facchinetti, "Digital twins in type 1 diabetes: A systematic review," *Journal of Diabetes Science and Technology*, 2024, pMID: 38887022.
- [56] T. Zhu, K. Li, P. Herrero, and P. Georgiou, "Population-specific glucose prediction in diabetes care with transformer-based deep learning on the edge," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 18, no. 1, pp. 145–156, 2024.
- [57] J. Han, Y. Wang, L. Liu, and D. Zhang, "Glu-ensemble: An ensemble deep learning framework for blood glucose forecasting in type 2 diabetes patients," *Heliyon*, vol. 10, no. 8, p. e29030, 2024.
- [58] Y. Li, X. Wang, M. Xu, and M. He, "Integrated image-based deep learning and language models for primary diabetes care," *Nature Medicine*, vol. 30, pp. 2419–2430, 2024.